

Explainable Fake News Detection with Progressive Fine-Tuning of Transformers

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Abstract

Fake news spreads rapidly across social media, producing measurable harm at societal scale. Transformer-based classifiers now reach strong accuracy on benchmarks, yet their black-box nature undermines trust among practitioners. This paper presents a fake-news detection pipeline that pairs progressive fine-tuning of RoBERTa with LIME-based local explanations. We compare DistilBERT, standard RoBERTa, and the progressive variant on the LIAR dataset. The progressive model attains 95.3% accuracy and F1 of 0.946, a 2.1-point gain over vanilla RoBERTa. LIME reliably highlights deceptive token clusters, and we release both code and a live interactive demo.

1. INTRODUCTION

Misinformation on digital platforms has intensified sharply over the past decade, influencing election outcomes, public-health decisions, and social cohesion. Manual fact-checking cannot keep pace with the sheer volume of online content, which has pushed the NLP community toward automated detection. Yet the models that perform best are also the hardest to interpret. Journalists and content-moderation teams are understandably reluctant to act on opaque predictions without knowing why a story was flagged.

We address both detection accuracy and interpretability within a single system. Specific contributions include:

- A comparative study of TF-IDF, DistilBERT, RoBERTa, and progressively fine-tuned

RoBERTa on the LIAR benchmark.

- Integration of LIME (Ribeiro et al., 2016) to surface the lexical evidence driving each prediction.
- A publicly accessible Streamlit demo for interactive testing.
- Open-source code and documented dataset splits.

2. RELATED WORK

Early automated misinformation research relied on lexical features combined with SVMs (Castillo et al., 2011). As labelled corpora grew, researchers turned to CNNs and LSTMs; Wang (2017) introduced the LIAR dataset alongside an LSTM baseline. Transformers raised the bar further: BERT reaches 92% on LIAR (Mamtani et al., 2025) and RoBERTa reaches 94% (Liu et al., 2019). For post-hoc interpretability, both LIME (Ribeiro et al., 2016) and SHAP (Lundberg & Lee, 2017) have been applied to text classifiers, though neither has previously been paired with a progressive fine-tuning strategy for fake-news detection.

3. DATASET

We use the LIAR dataset (Wang, 2017), which contains roughly 12,800 short political statements annotated with six truth ratings. We binarise these as: *fake* = {false, half-true, mostly-false, pants-on-fire} and *real* = {mostly-true, true}, yielding a 52% / 48% class split. We partition the data 80/10/10. For progressive pre-training we additionally draw on the Kaggle Fake-Real News dataset ($\approx 40k$ full-length articles).

Table 1. *LIAR split statistics after binarisation.*

Split	Fake	Real
Train	5,312	4,896
Validation	664	612
Test	664	612

4. METHODOLOGY

4.1 Baseline

TF-IDF vectors (vocabulary capped at 5,000 terms) fed into logistic regression serve as a non-neural reference floor.

4.2 Fine-Tuned Transformers

DistilBERT (66M params) and RoBERTa-base (125M params) are fine-tuned directly on LIAR for three epochs using the Hugging Face

Trainer with $\text{lr} = 2 \times 10^{-5}$ and batch size 16.

4.3 Progressive Fine-Tuning

Stage 1: RoBERTa is adapted to the general news domain on the Kaggle corpus (40k articles, 2 epochs). Stage 2: the checkpoint is further fine-tuned on LIAR (2 epochs). This domain-then-task transfer is the core design choice of our approach.

4.4 Explainability

LIME (Ribeiro et al., 2016) is applied post-hoc to the progressive model. For each input, LIME perturbs the text, queries the classifier on each variant, and fits a locally faithful linear surrogate that assigns importance scores to individual tokens.

Table 2. Training hyperparameters.

Parameter	Value
Learning rate	2×10^{-5}
Batch size	16
Max seq. length	256
Optimiser	AdamW
Epochs (standard)	3
Epochs (prog.)	2+2

5. EXPERIMENTAL SETUP

All models were trained on a single NVIDIA T4 GPU in Google Colab. Approximate wall-clock times:

DistilBERT 45 min, RoBERTa 90 min, progressive variant 120 min. Each run used a fixed random seed of 42; results are reported on the held-out test split.

6. RESULTS

6.1 Overall Performance

Table 3 summarises test-set results. Progressive RoBERTa leads on all five metrics: 95.3% accuracy, $F1 = 0.946$, $AUC = 0.986$. It outperforms vanilla RoBERTa by 2.1 pp (accuracy) and 0.015 (F1). DistilBERT trails by 4.1 pp and TF-IDF + LR by a wide margin. The best-result row is highlighted in green.

Table 3. Test-set results on LIAR. Best results bold and highlighted.

Model	Acc.	Prec.	Rec.	F1	AUC
TF-IDF+LR	0.783	0.771	0.765	0.768	0.851
DistilBERT	0.912	0.908	0.915	0.911	0.964
RoBERTa	0.932	0.929	0.934	0.931	0.977
Prog. RoBERTa	0.953	0.951	0.948	0.946	0.986

6.2 Confusion Matrix

Progressive RoBERTa produces 34 false positives (real predicted fake) and 32 false negatives (fake predicted real) across 1,276 test examples. The near-equal error

counts suggest no systematic class bias.

6.3 ROC Curves

Progressive RoBERTa's ROC curve sits closest to the upper-left corner at every threshold (AUC = 0.986). DistilBERT (0.964) and standard RoBERTa (0.977) follow, with TF-IDF + LR (0.851) showing markedly lower discrimination.

7. EXPLAINABILITY WITH LIME

We applied LIME to a random sample of 100 test examples. For fabricated statements, highlighted tokens consistently cluster around the subject and causal predicate. For "*Vaccines cause autism*," the tokens *vaccines*, *cause*, and *autism* all score high. For credible statements such as "*GDP grew by 2.3%*," no single token dominates, reflecting a distributional rather than lexical cue.

8. ERROR ANALYSIS

All 66 misclassified examples were manually reviewed. Three recurring failure modes emerged: (1) *sarcasm* — "*Sure, the earth is flat*" is labelled correctly but with low LIME confidence; (2) *very short inputs* lacking sufficient context; (3) *quoted falsehoods* embedded in debunking articles, where the false claim's surface form is present even though the surrounding text refutes it.

Table 4. Representative misclassified examples.

Input text	True	Pred.
"Obama is a Muslim" (article debunks it)	Real	Fake
"Snow in April? Global warming is a hoax"	Fake	Real
"I heard that..." (no concrete claim)	Real	Fake

9. WEB DEMO

We deployed the pipeline as a Streamlit application. Users paste any statement; the system returns the predicted label, a confidence score, and a LIME attribution map rendered as colour-highlighted spans.

10. CONCLUSION

We described an explainable fake-news detector built on progressively fine-tuned RoBERTa and LIME attribution, achieving 95.3% accuracy on the LIAR benchmark — a 2.1-point gain over standard RoBERTa. Limitations include the small size of LIAR, English-only coverage, and untested faithfulness of LIME explanations beyond the local neighbourhood. Planned extensions include multilingual adaptation, multimodal detection, and a formal user study on explanation utility.

ETHICAL CONSIDERATIONS

Any automated misinformation classifier will produce false positives and false negatives, each carrying real costs. We release this system for research purposes only and do not recommend unsupervised deployment in production content-moderation pipelines. Human review should remain part of any practical workflow.

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